MIL-S-19500/193C(EL)
15 September 1971
SUPERSEDING
MIL-S-19500/193B
19 April 1968
(See 6.3)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER

TYPES 1N457, 1N458, AND 1N459

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon diode rectifier and shall be in accordance with MIL-S-19500 except as otherwise specified herein.

1.2 Physical dimensions. See figure 1 (DO-7).

1.3 Ratings and characteristics.

Type	v_r	V_{RM} (wkg)	I_o 1/ T_A = 25° C	i_f
	v(pk)	v(pk)	mAdc	ma
1N457	70	60	75	225
1N458	150	125	55	165
1N459	200	175	40	120

1/ Derate I_o linearly to 0.0 mAdc at 150° C.

OPERATING CASE TEMPERATURE: -65° to +150° C

STORAGE TEMPERATURE: -65° to +200° C

BAROMETRIC PRESSURE REDUCED: 15 mm Hg

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Diodes shall be of the design, construction, and physical dimensions specified in figure 1.

* 3.3.1 Lead finish. Lead finish shall be gold plated or tinned. Lead finish may be specified in the contract or order (see 6.2) without affecting the qualified product status of the device or applicable JAN marking.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Manufacturer's identification.

(b) Country of origin.

(c) Acceptance data and inspection lot identification.

3.5.1 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in Groups A, B, and C.

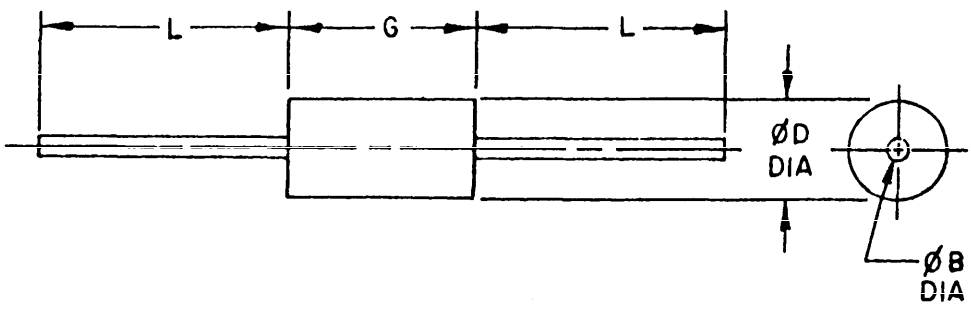
4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot, and thereafter every six months during production.

* 4.3.4 Thermal shock (temperature cycling). Samples that have been subjected to the group B thermal shock test may be continued on test to 25 cycles in order to satisfy group C thermal shock test requirements. These samples shall be predesignated, and shall remain subjected to the group C 25 cycle acceptance evaluation after they have passed the group B 10 cycle acceptance criteria. The cumulative total of failures found during the 10 cycle test and during the subsequent interval up to 25 cycles shall be computed for 25 cycle acceptance criteria.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
G	.230	.300	5.84	7.62	
ØD	.085	.130	2.16	3.30	2
L	1.000	1.500	25.40	38.10	
ØB	.018	.022	.46	.56	1

NOTES:
1. The specified lead diameter applies in the zone between .050 (1.27 mm) and 1.000(25.40 mm) from the diode body. Outside of this zone the lead diameter is not controlled.
2. The minimum body diameter shall be maintained over .15(.38 mm) inch of body length.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Semiconductor device, diode, types 1N457, 1N458, and 1N459 (DO-7).

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4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

* 4.4.1 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine wave form of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

* 4.4.2 Resistance to solvents. The devices shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the device body where marking has been applied shall be brushed.
(b) After completion of the test, there shall be no evidence of mechanical damage to the device, and markings shall have remained legible.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2							
Forward voltage:	4011		5				
1N457		I_F = 20 mAdc		V_F	---	1.0	Vdc
1N458		I_F = 7 mAdc		V_F	---	1.0	Vdc
1N459		I_F = 3 mAdc		V_F	---	1.0	Vdc
Reverse current:	4016	DC method					
1N457		V_R = 60 Vdc		I_R	---	25	nAdc
1N458		V_R = 125 Vdc		I_R	---	25	nAdc
1N459		V_R = 175 Vdc		I_R	---	25	nAdc
Peak reverse current at peak reverse voltage:	4016	AC method; f = 60 Hz					
1N457		v_r = 70 v(pk)		i_r	---	100	μa(pk)
1N458		v_r = 150 v(pk)		i_r	---	100	μa(pk)
1N459		v_r = 200 v(pk)		i_r	---	100	μa(pk)
High-temperature operation:		T_A = 150°C					
Reverse current	4016	DC method					
1N457		V_R = 60 Vdc		I_R	---	5	μAdc
1N458		V_R = 125 Vdc		I_R	---	5	μAdc
1N459		V_R = 175 Vdc		I_R	---	5	μAdc

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026	Immersion depth to within 0.10 inch of body		---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition F		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Terminal strength: Tension	2036	Test condition A; weight = 2 lb; t = 30 ±3 sec		---	---	---	---
*Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. D or E for gross leaks		---	---	---	---
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Forward voltage:	4011						
1N457		I_F = 20 mAdc		V_F	---	1.0	Vdc
1N458		I_F = 7 mAdc		V_F	---	1.0	Vdc
1N459		I_F = 3 mAdc		V_F	---	1.0	Vdc
Reverse current:	4016	DC method					
1N457		V_R = 60 Vdc		I_R	---	25	nAdc
1N458		V_R = 125 Vdc		I_R	---	25	nAdc
1N459		V_R = 175 Vdc		I_R	---	25	nAdc
Subgroup 3			15				
Shock	2016	Nonoperating; 500 G; t = 1.0 msec; 5 blows each orientation X_1, Y_1, and Y_2		---	---	---	---
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	Nonoperating; 20,000 G; X_1, Y_1, and Y_2 orientations		---	---	---	---
End points: (Same as for subgroup 2)							
Subgroup 4			20				
Terminal strength: Lead fatigue	2036	Test condition E		---	---	---	---
End points: (Same as for subgroup 2)							

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 5			λ = 5				
High-temperature life (nonoperating)	1031	T_A = 200° C		---	---	---	---
End points:							
Forward voltage:	4011						
1N457		I_F = 20 mAdc		V_F	---	1.0	Vdc
1N458		I_F = 7 mAdc		V_F	---	1.0	Vdc
1N459		I_F = 3 mAdc		VF	---	1.0	Vdc
Reverse current:	4016	DC method					
1N457		V_R = 60 Vdc		I_R	---	50	nAdc
1N458		V_R = 125 Vdc		I_R	---	50	nAdc
1N459		V_R = 175 Vdc		I_R	---	50	nAdc
Subgroup 6			λ = 5				
Steady-state operation life: (See 4.4.1)	1026	f = 60 Hz					
1N457		I_O = 75 mAdc; v_r = 60 v(pk)		---	---	---	---
1N458		I_O = 55 mAdc; v_r = 125 v(pk)		---	---	---	---
1N459		I_O = 40 mAdc; v_r = 175 v(pk)		---	---	---	---
End points: (Same as for subgroup 5)							
Subgroup 7			15				
Low-temperature operation:		T_A = -65° C					
Reverse current:	4016						
1N457		V_R = 60 Vdc		I_R	---	25	nAdc
1N458		V_R = 125 Vdc		I_R	---	25	nAdc
1N459		V_R = 175 Vdc		I_R	---	25	nAdc

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TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			15				
Barometric pressure, reduced (altitude operation)	1001	15 mm Hg; t = 60 sec		---	---	---	---
Measurement during test:							
Reverse current:	4016	DC method		I_R			
1N457		V_R = 60 Vdc			---	25	nAdc
1N458		V_R = 125 Vdc			---	25	nAdc
1N459		V_R = 175 Vdc			---	25	nAdc
Subgroup 2			15				
Salt atmosphere (corrosion)	1041			---	---	---	---

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

* 6.2 Ordering data. Lead finish, if other than tinned (see 3.3.1).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

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SPECIFICATION MIL-S-19500/193C(EL) SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER TYPES 1N457,1N458, AND 1N459

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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